

CLINIC MANAGEMENT SYSTEM

UNIT: DESIGN AND USE A DATABASE

UNIT CODE: ICT/OS/CP/CR/o4/6/A

PROJECT

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TASK 1 : Set up a Database System

◆ Compare RDBMS Tools

- i. Comparison of Three RDBMS tools based on performance, scalability, cost, ease of use and compatibility with the system requirements.

1. Performance

Feature	MySQL	PostgreSQL	SQL Server
Read Performance	Excellent for read-heavy operations	Good, slightly slower on simple reads	High performance with optimized indexing
Write Performance	Fast, but less strict on ACID	Slower but more ACID-compliant	Very fast for both OLTP and OLAP workloads
Complex Queries	Limited advanced features	Strong support for complex queries	Strong support, integrated with analytic

2. Scalability

Feature	MySQL	PostgreSQL	SQL Server
Vertical Scaling	Good	Good	Excellent (with Enterprise features)
Horizontal Scaling	Limited native support (via clustering)	Better via tools like Citus	Advanced support in Azure SQL Database
Cloud Integration	Fully supported (AWS RDS, GCP)	Fully supported	Native Azure support

3. Cost

Feature	MySQL	PostgreSQL	SQL Server
Licensing	Free (Community), Paid (Enterprise)	Completely free	Expensive for Enterprise; free Express edition
Hosting Cost	Low to medium	Low to medium	High for full features
Total Cost	Low	Very low	High in corporate environments

4. Ease of Use

Feature	MySQL	PostgreSQL	SQL Server
Setup & Installation	Simple, fast setup	Moderate complexity	Easy with GUI-based setup
GUI Tools	MySQL Workbench	pgAdmin	SQL Server Management Studio (SSMS)
Learning Curve	Low to medium	Medium (more complex features)	Low (especially for Windows users)

5. Compatibility with System Requirements

Feature	MySQL	PostgreSQL	SQL Server
OS Support	Windows, Linux, macOS	Windows, Linux, macOS	Primarily Windows; Linux supported (limited)
Integration Options	Web apps, PHP, Java, Python	Strong integration, especially with Python	Best with .NET, Power BI, Azure services
Hardware Requirements	Lightweight	Slightly heavier than MySQL	Heavy (especially on Windows environments)

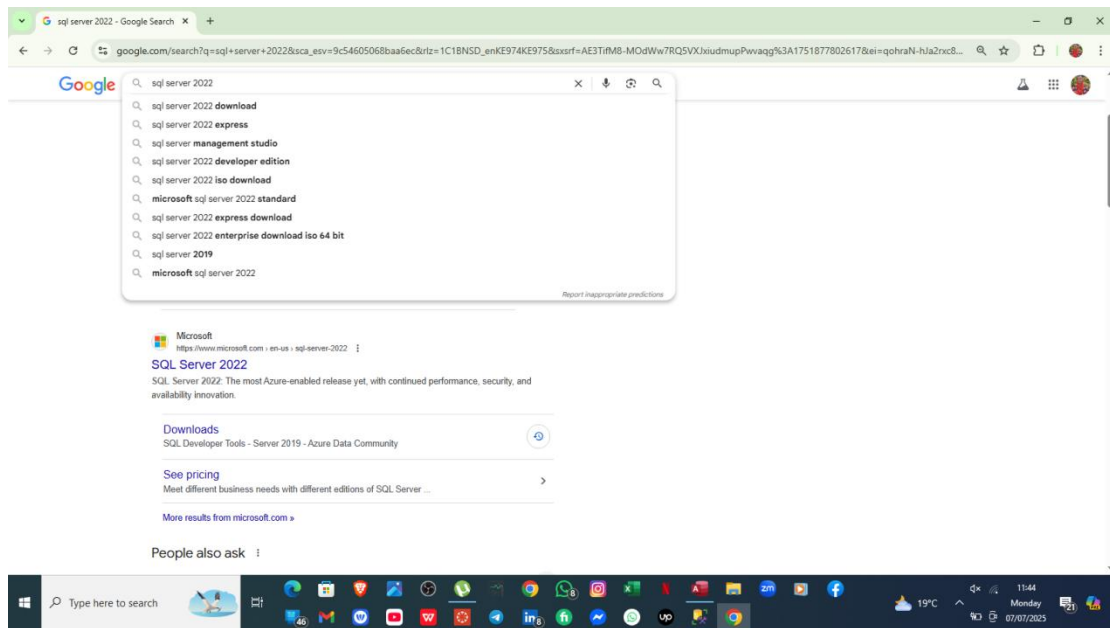
ii. Table Summarizing the comparison and a recommendation for the clinic.

Feature	Comparison	Recommendation
Performance	Excellent for read-heavy operations	MySQL
Scalability	Excellent (with Enterprise features)	SQL Server
Cost	Free (Community), Paid (Enterprise)	MySQL
Ease of use	Simple, and fast	MySQL
Compatibility with System Requirements	Windows, Linux, OS	MySQL

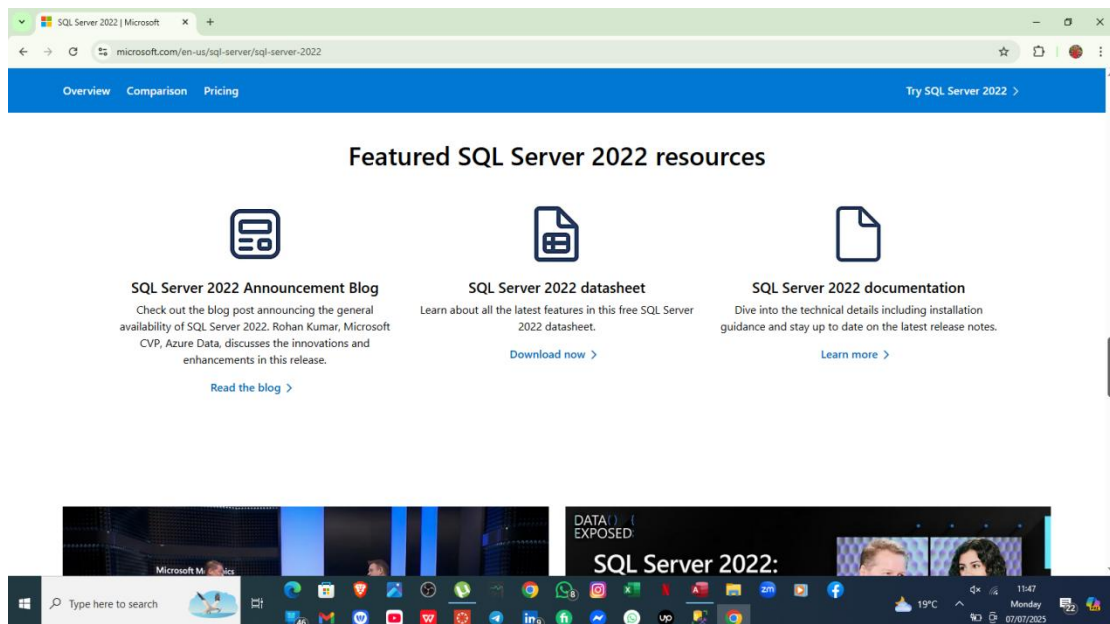
◆ Installing a database and/or Web Server.

✓ MySQL

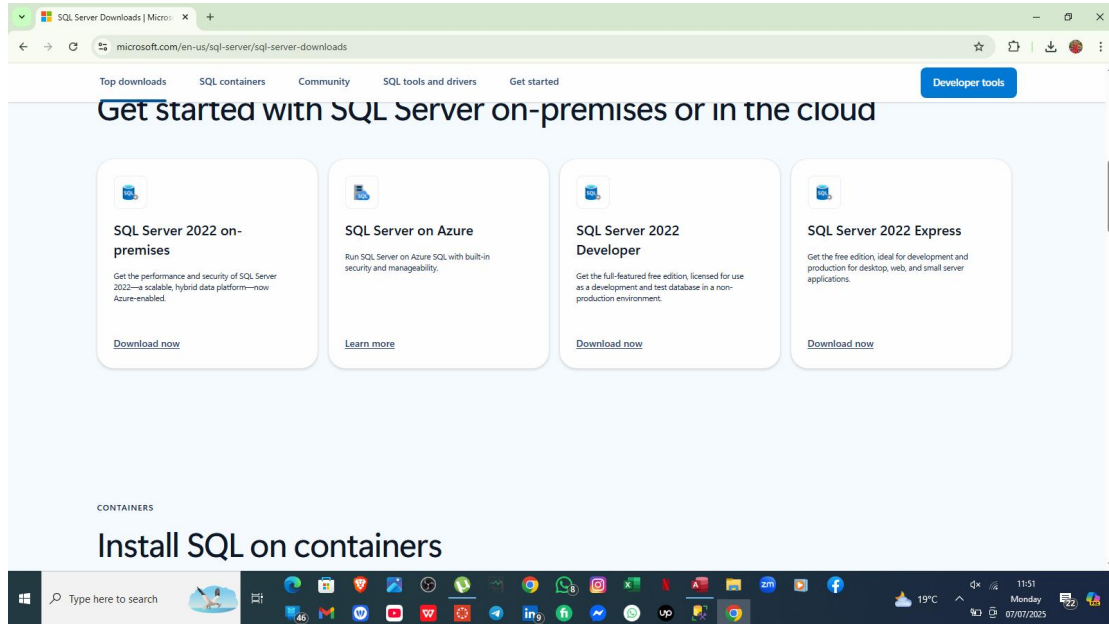
i. Open a web browser and search for SQL Server download.



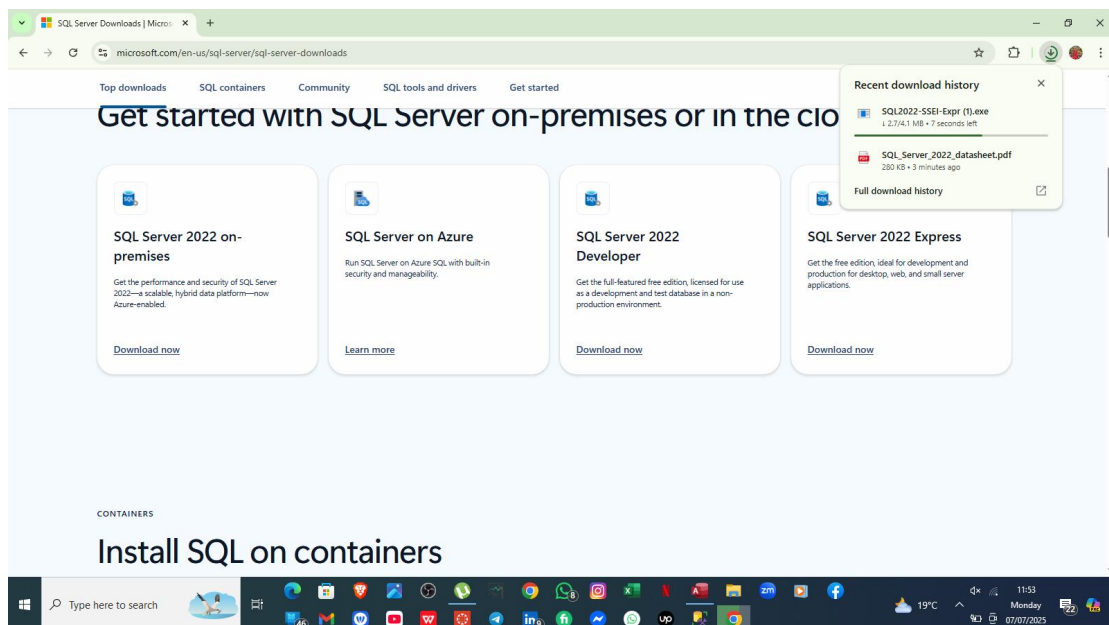
ii. Click on SQL Server 2022 on Microsoft.



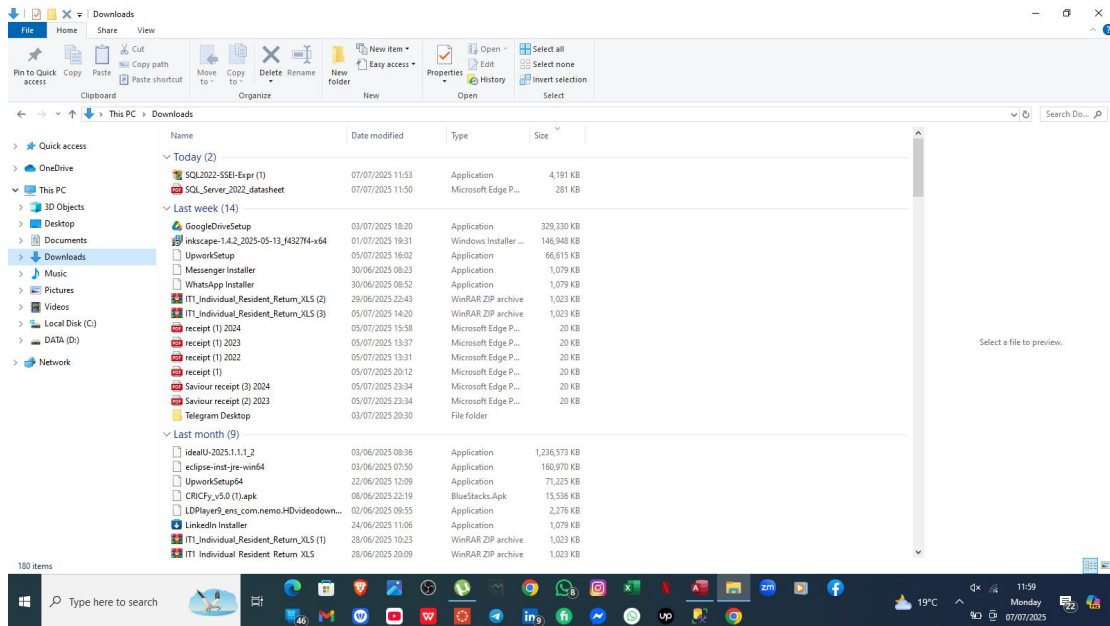
iii. After opening, scroll down and select SQL Server 2022 Express



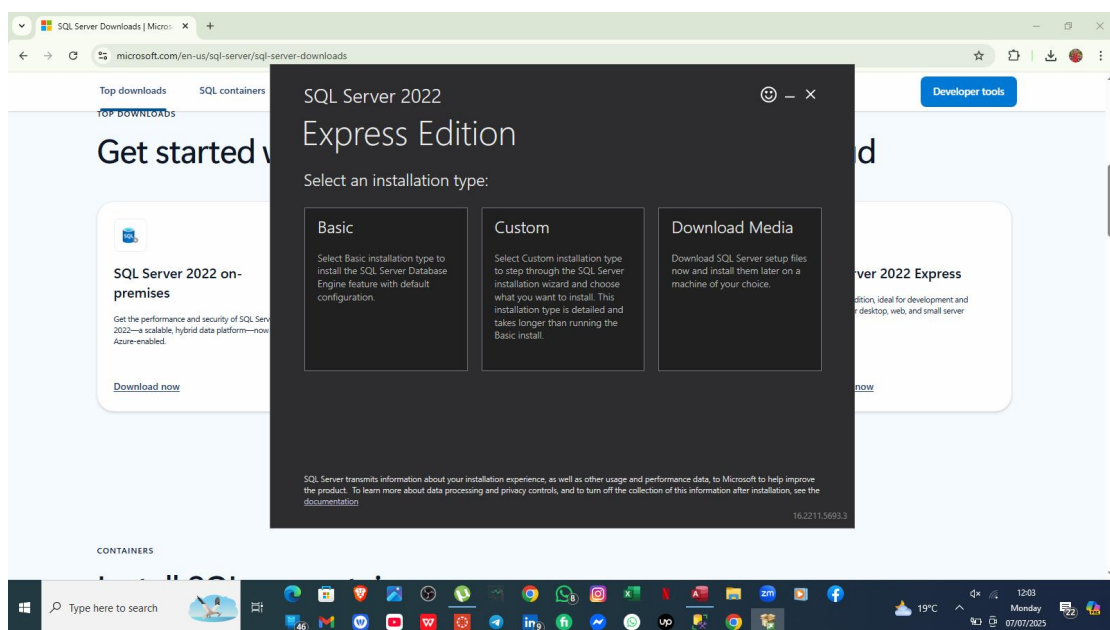
iv. Click download and observe the downloading progress on the downloads tab.



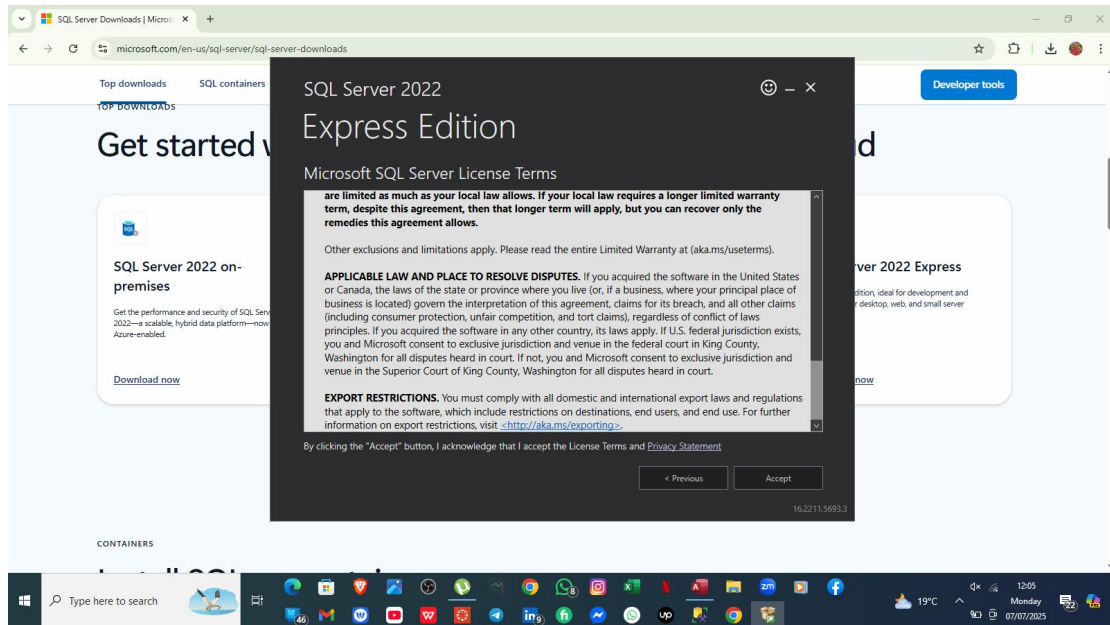
- v. After finishing the download, open file explorer, open my downloads where the server is located.



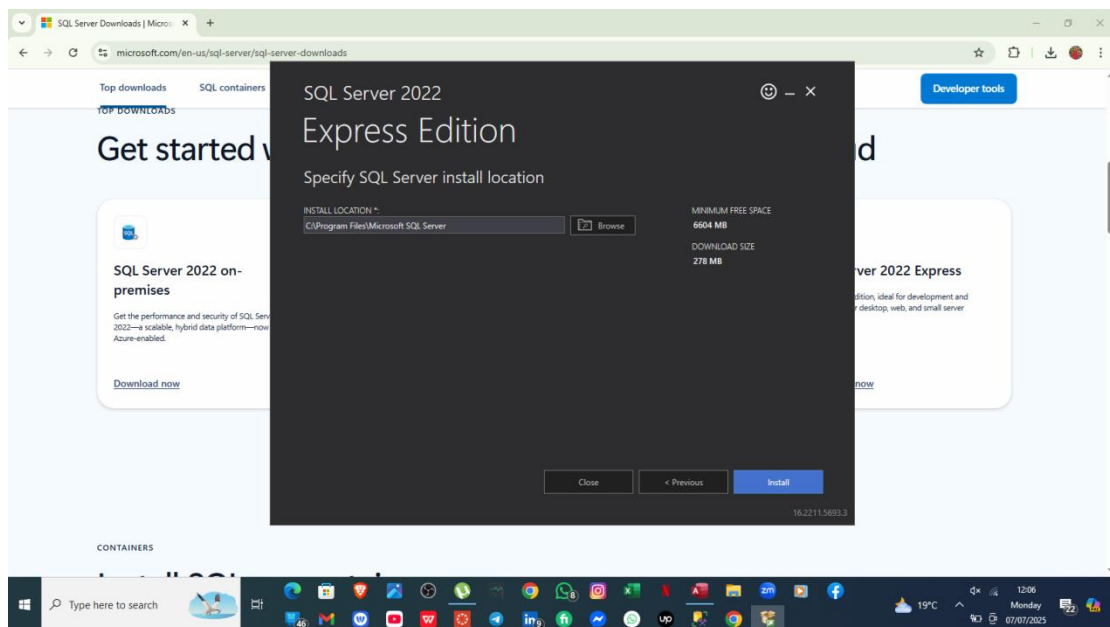
- vi. Click the server to Run, and allow it to make changes on PC by clicking YES.
- vii. Select the Basic to download it, and accept and agree with terms and conditions by clicking agree.



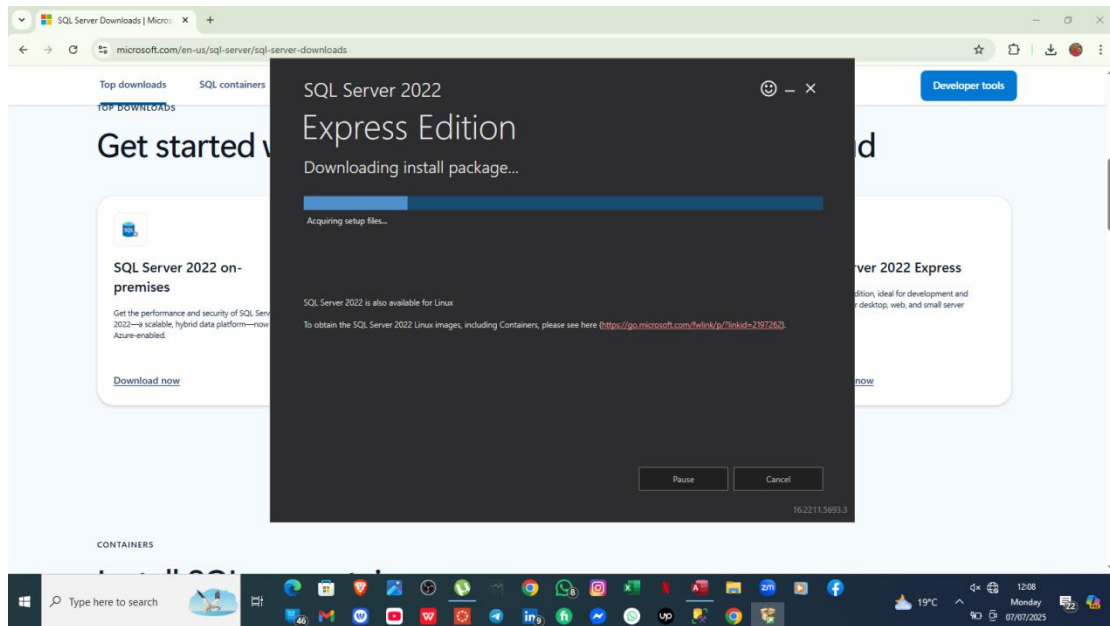
viii. Click Accept



ix. Click Install to start the installation process.



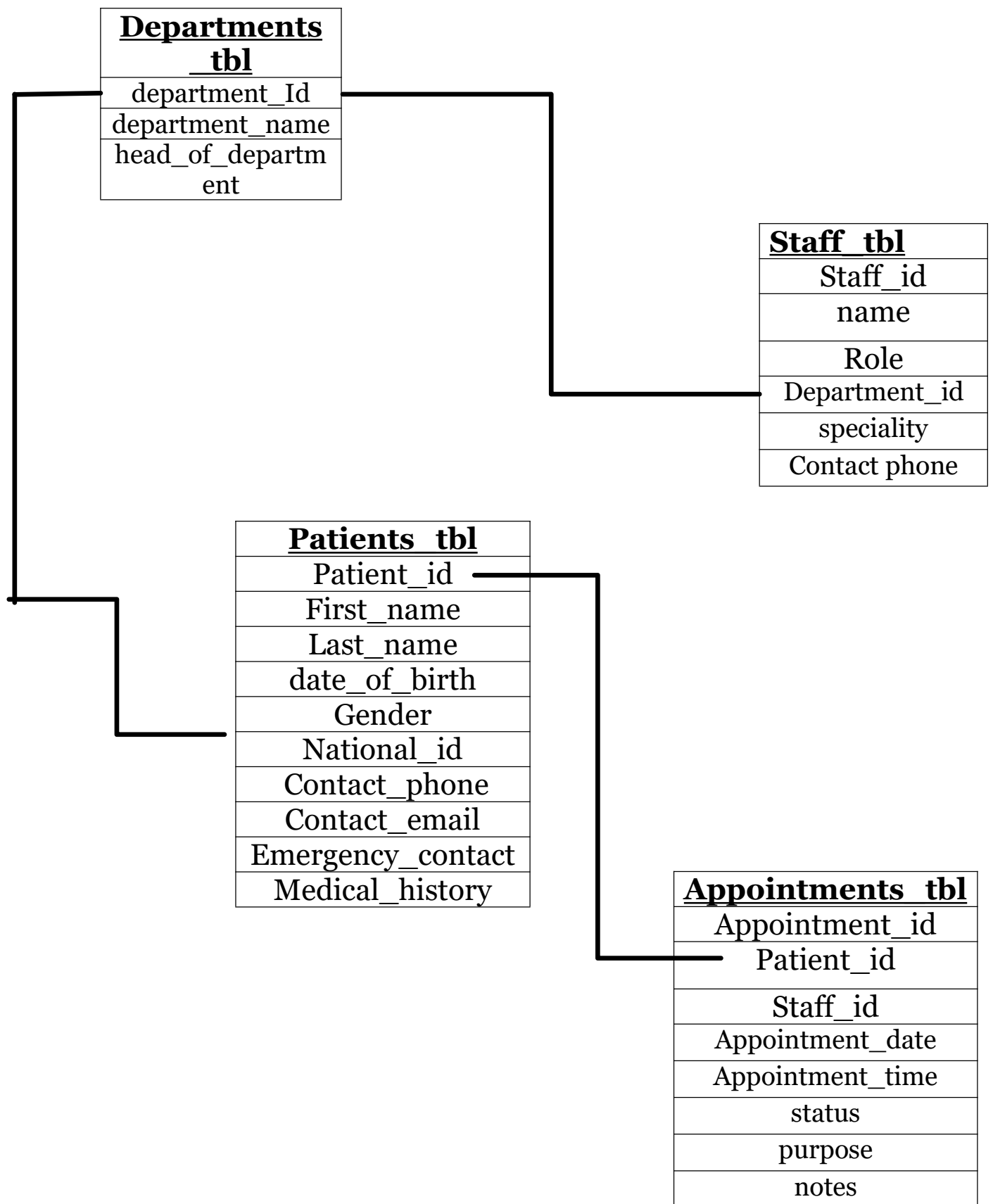
- x. Give it time to finish installation, and make sure that all the packages and SSMS are downloaded successfully.



Entities and Relationships

Entity A	Relationship Type	Entity B	Rationale/Cardinality
Patients	Books	Appointments	1-to-Many (One patient can have many appointments)
Staff	Handles	Appointments	1-to-Many (One staff member, usually a doctor handles many appointments)
Appointments	Generates	Bills	1-to-1 (Each appointment generates one bill)
Bills	Includes	BillItems	1-to-Many (One bill can include multiple bill items)
MedicalServices	Defines	BillItems	1-to-Many (One service can appear on many bill items)
Appointments	Results in	Prescriptions	1-to-Many (One appointment can result in zero or many prescriptions)
Patients	Receives	Prescriptions	1-to-Many (One patient can receive many prescriptions)
Staff	Issues	Prescriptions	1-to-Many (One doctor issues many prescriptions)
Departments	Employs	Staff	1-to-Many (One department employs many staff members)

ERD DIAGRAM



Crude Matrix

<u>Table</u>	<u>Admin (Full Access)</u>	<u>Doctor</u>	<u>Nurse</u>	<u>Receptionist</u>
Departments	CRUD	R	R	R
Staff	CRUD	R	R	R
MedicalServices	CRUD	R	R	R
Patients	CRUD	R, U	R	R
Appointments	CRUD	R, C, U	R, C, U	CRUD
Prescriptions	CRUD	R, C	R, C	R
Bills	CRUD	R	R	R
BillItems	CRUD	R	R	R

KEY:

C: Create (INSERT)

R: Read (SELECT)

U: Update (UPDATE)

D: Delete (DELETE)

Compiling Project

Commenting all SQL scripts to Explain the purpose of each statement.

SQL Statement	Purpose
CREATE TABLE	Syntax used to create a table in a database.
INSERT INTO	Used to insert data into database tables.
SELECT	Used to retrieve the result of the stored data from a database table.
UPDATE	Used to update data in a table without having the need to delete it .
DELETE	Used to delete certain parts of data in a table without deleting the entire table.
DROP TABLE	Used to delete the entire table when it is no-longer needed.
ALTER TABLE	Used to alter the structure of a table, can Add, Rename, Modify or Change a column or Data Type in a table.
WHERE Clause	Used to filter rows from a table by applying a condition.
AND / OR Operators	Used to apply multiple conditions in the where clause.
LIKE Clause	Used to retrieve the values from a table that match a certain pattern using WHERE Clause.
ORDER BY Clause	Used to arrange the column values in a given / Specified order.
GROUP BY Clause	Used to group the values of a column together.
COUNT Function	Gives the number of Non-Null values present in the specified column.
AVG	Gives the average of Non-Null values present.
CREATE VIEW	Used to create views in a database tables.

DROP VIEW	Used to delete the entire view created in a table.
JOIN	Used to join data from different tables in a database.